

# Sketch2Radiation: Predicting annual solar radiation maps of buildings from architectural sketches with depth and camera pose

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## ABSTRACT

Early-stage architectural design often lacks timely feedback on annual facade-level solar radiation, as conventional simulation tools require complete 3D models and time-consuming simulation, slowing design iteration. To address this gap, we propose Sketch2Radiation, a view-aligned surrogate that directly predicts annual solar radiation on visible building surfaces from a single-view digitally generated sketch, conditioned on the corresponding depth (Z-buffer) map, camera pose, and climate descriptor. In contrast to conventional geometry-to-performance surrogates that require complete 3D representations and often produce non-view-aligned outputs, Sketch2Radiation directly maps the designer's current view to a view-aligned facade radiation map for rapid, in-workflow interpretation and iteration. We release a large-scale (480,000 samples) multi-climate dataset spanning 1000 buildings  $\times$  24 views  $\times$  20 cities, with annual radiation maps generated from high-fidelity Radiance-based simulation. Experiments in both single-city and multi-city settings show stable and geometry-consistent predictions. In the multi-city experiment, the model achieves MSE = 0.000236, MAE = 0.002371, SSIM = 0.990219, FSIM = 0.995537 and PSNR = 39.20 dB, indicating strong agreement with simulation outputs. Design-oriented PV indicators also remain accurate, with an average error below 1% in total solar potential prediction, enabling reliable PV site-selection decisions during design. Rather than replacing detailed simulation, Sketch2Radiation serves as an early-stage screening tool for concept-phase massing and facade design, lowering the barrier to incorporating solar considerations into routine design workflows. Future works include SVF-based or lightweight context-aware calibration, and extend to freehand pen-and-paper sketches.

## List of Abbreviations

|          |                                   |
|----------|-----------------------------------|
| 2AFC     | Two-Alternative Forced Choice     |
| AdaBoost | Adaptive Boosting                 |
| ANN      | Artificial Neural Network         |
| BIPV     | Building-integrated Photovoltaics |
| CNN      | Convolutional Neural Network      |
| DF       | Daylight Factor                   |
| DGP      | Daylight Glare Probability        |
| DHI      | Diffuse Horizontal Irradiance     |
| DNI      | Direct Normal Irradiance          |
| DT       | Decision Tree                     |
| FSIM     | Feature Similarity Index Measure  |
| GAN      | Generative Adversarial Network    |
| GAT      | Graph Attention Network           |
| GBM      | Gradient Boosting Machine         |
| MLP      | Multi-layer Perceptron            |

|        |                                     |
|--------|-------------------------------------|
| PINNs  | Physics-Informed Neural Networks    |
| PSNR   | Peak Signal-to-Noise Ratio          |
| PSO    | Particle Swarm Optimization         |
| ResNet | Residual Network                    |
| RF     | Random Forest                       |
| sDA    | Spatial Daylight Autonomy           |
| SLP    | Single-Layer Perceptron             |
| SSIM   | Structural Similarity Index Measure |
| SVF    | Sky View Factor                     |
| TCN    | Temporal Convolutional Network      |
| UDI    | Useful Daylight Illuminance         |
| XGB    | Extreme Gradient Boosting           |

## 1. Introduction

Buildings and construction are responsible for roughly 37% of global energy-related CO<sub>2</sub> emissions [1], making early design decisions a

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decisive factor in shaping the long-term carbon trajectory of the built environment [2]. Building-integrated photovoltaics (BIPV) are widely regarded as a key pathway toward net-zero buildings, embedding on-site energy generation within the building envelope while satisfying architectural and functional requirements. However, the effective yield of BIPV is highly sensitive to massing, orientation, self-shading and local climate conditions [3–8]. Without a quantitative characterization of the annual solar radiation distribution over building surfaces at the conceptual stage, designers lack a reliable basis for distinguishing genuinely PV-suitable facades from marginal ones, and for assessing how geometric operations alter long-term solar potential [9–12]. As a result, conceptual design demands a workflow that is rapid, climate-aware, and compatible with frequent modification.

Current building performance simulation (BPS) tools, including Radiance-based simulation tools (e.g., Ladybug Tools, Daysim, Sefaira and ClimateStudio), achieve high predictive accuracy but remain poorly aligned with the workflow of conceptual design. They typically require clear 3D geometry, detailed material definitions and well-defined boundary conditions, with runtimes commonly extending to minutes or longer. Such requirements are difficult to satisfy during massing exploration and facade iteration and tend to disrupt the continuous cycle of “ideation-adjustment-feedback”. At this stage, designers primarily seek immediate directional insight rather than engineering-grade precision [13,14]. Despite this need, there remains no method that leverages the lightweight visual information already generated during design work to provide rapid yet physically meaningful predictions of annual solar radiation.

From the perspective of design methodology, even as CAD and BIM environments continue to evolve [15], sketches and simplified massing models remain the primary working language of conceptual design [16–19]. Meanwhile, modern modeling tools automatically produce some signals during every viewport operation that are physically relevant, such as sketches, depth (Z-buffer) map [20] and camera parameters. These view-aligned signals are lightweight but accurately reflect visible surfaces and geometric relationships, forming an input representation that aligns seamlessly with early-stage design workflows. This motivates the central question of the study: Can annual solar radiation maps on visible building surfaces be predicted directly from the lightweight signals (such as sketches, depth maps, camera poses) already available during the conceptual modeling process?

To address this gap, this study proposes Sketch2Radiation (Fig. 1), a view-aligned surrogate model for predicting annual solar radiation from lightweight signals generated during conceptual design. The model operates on representations that are readily available during standard modeling workflows, including sketches, depth maps, and camera parameters. By processing these view-aligned signals, the approach avoids

reliance on fully specified 3D geometry and enables direct prediction within the design view.

Conventional geometry-to-performance surrogates are formulated as mappings  $f: G \rightarrow P$ , where  $G$  denotes explicit 3D geometry and  $P$  denotes performance quantities defined in object or world coordinates. In contrast, Sketch2Radiation is formulated as a mapping  $g: (V_s, V_d, p, c) \rightarrow R_v$ , where the inputs consist of view-aligned signals, including sketch  $V_s$ , depth  $V_d$ , camera pose  $p$ , and climate descriptor  $c$ , and the output  $R_v$  is a radiation field defined directly in the same view coordinate system. This formulation eliminates intermediate steps such as geometry reconstruction and reprojection while preserving geometric cues through depth and perspective. This view-aligned formulation allows performance prediction to be performed directly within the design view, maintaining consistency between input representation and output field. As a result, the prediction process becomes naturally integrated with viewing and editing operations during conceptual design.

This study makes four main contributions. (i) it introduces a view-aligned formulation for solar radiation prediction that directly operates on design-native visual representations. (ii) it proposes a surrogate modeling framework that enables rapid, climate-aware estimation of facade solar radiation during conceptual design. (iii) it establishes a large-scale, multi-view, cross-climate dataset to support the development and benchmarking of such models. (iv) it presents a comprehensive evaluation framework that assesses physical consistency, perceptual fidelity, and design utility, thereby clarifying the applicability and limitations of the proposed approach.

## 2. Related work

### 2.1. Sketch-based surrogates

Sketches [17–19] play a central role in early-stage architectural design and have been widely explored as inputs for automated geometry generation. Recent sketch-based surrogate research has shown that reconstructing 3D geometry from 2D line drawings is technically feasible. Representative methods include Floor-SP [21] and Plan2Scene [22], which generate room layouts and floorplans from sketches; Sketch2CAD [23] and Sketch2Mesh [24], which reconstruct CAD-like volumes or coarse massing from drawings; and Sketching Reality [25], which converts freehand architectural sketches into textured 2.5D models. Vitruvio [26] further supports view-based reconstruction of 3D meshes while preserving orientation-sensitive detail.

A few studies have begun to use sketches as inputs to physical surrogates rather than pure geometry. Sketch2Stress [27], for example, predicts structural stress distributions and normal maps directly from 2D sketches with user-specified force points, illustrating that it is possible to

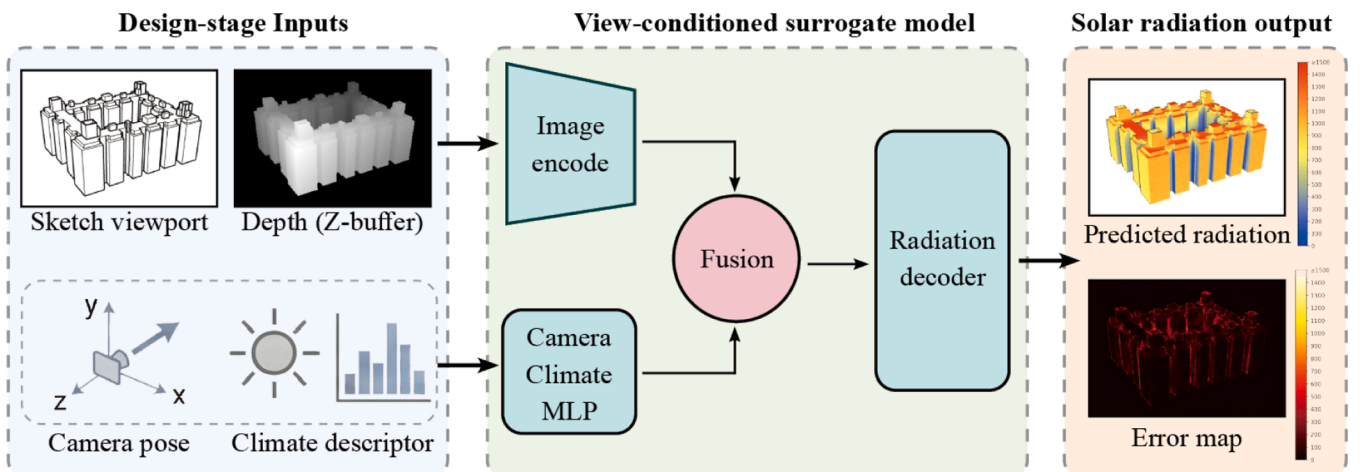


Fig. 1. The workflow of Sketch2Radiation: A view-aligned surrogate model of the annual solar radiation distribution for early architectural design.

map sketch space to physical fields without first recovering full geometry. However, this line of research remains largely confined to structural mechanics and low-dimensional targets (e.g., local stress or surface normals) and has yet to be extended to climate-driven building environmental fields that couple geometry with weather, such as annual solar radiation.

Overall, most sketch-based methods treat sketches mainly as a front-end modality for generating or refining 3D models, while environmental performance assessment is deferred to downstream simulation tools. A key gap is the lack of a framework that uses the same view representations architects work with, such as sketch-like images, depth maps, and camera poses, as inputs and directly outputs facade-level performance fields.

## 2.2. Image/grid-based building performance surrogates

Meanwhile, image-to-image translation and grid-to-field prediction have made rapid progress in the fields of computer vision and building performance. From Pix2Pix [53] and CycleGAN [54] to SPADE [55], Palette [56], diffusion models [57,58] and ControlNet [59], many methods can generate visually plausible images conditioned on structural inputs such as edge maps and semantic layouts. These models, however, are predominantly optimized for perceptual quality rather than physical fidelity, which limits their direct applicability to long-term environmental quantities such as annual solar radiation. Within the building performance community, a growing body of work has developed image/grid-to-field surrogates for indoor daylight and related metrics (Table 1). Methodologically, these studies can be grouped into three main types:

**Data-driven CNN/GAN regression models.** He et al. (2021) [28] use ResNet50 and pix2pix to encode floorplans as distance-to-window images and predict daylight metrics and indoor illuminance distributions. Han et al. (2021) [31] and Le-Thanh et al. (2022) [32] employ feed-forward ANNs to predict point-wise UDI/DA and their spatial distributions.

**Physics-informed neural networks (PINNs).** Labib et al. (2025) [37] embed radiative transfer and diffusion approximations into PINN loss terms, learning daylight autonomy (DA) fields from a limited set of Daysim simulations and integrating the trained models into Grasshopper for instantaneous feedback.

**Multimodal generative frameworks.** Li et al. (2024, 2025) [35,38] and Díez et al. (2025) [36] use multimodal pix2pix-GAN or GAN-CNN frameworks to map solid-angle or semantic grids and climate vectors to building illuminance or luminance maps and daylight factor (DF) fields.

Collectively, these studies demonstrate that CNN-, GAN-, and PINN-based surrogates can predict indoor lighting fields (UDI, sDA, DGP, etc.) with sufficient accuracy to replace Radiance-based or DAYSIM simulations at millisecond-to-second timescales, greatly accelerating daylight evaluation in design tools. However, existing image/grid-to-field surrogates focus almost exclusively on interior lighting, whereas facade-level annual solar radiation distributions on building exteriors remain largely unexplored. In addition, most approaches assume that clean geometry is already available in the form of floorplans, complete 3D models, or highly structured grids, and therefore do not operate in the view coordinate system that architects actually use during conceptual design. As a result, there is still no alternative method or surrogate that can directly use view-based information as input to output an annual solar radiation distribution map within the same view. Sketch2Radiation is designed to fill this gap.

## 2.3. Solar radiation surrogates

For solar radiation assessment and BIPV design, a number of data-

**Table 1**

Summary of recent research on physics-informed and data-driven surrogate models for predicting building daylight factors, solar radiation, and energy.

| Ref  | Objective | Methods                        | Inputs                                                                                | Outputs                                                                           |
|------|-----------|--------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| [28] | Daylight  | ResNet50 CNN, Pix2Pix          | Grayscale floorplan images                                                            | Static annual daylight metrics, grid-based indoor illuminance distribution images |
| [29] | Daylight  | ANN                            | Intermediary features and facade geometry                                             | Grid-level DA and ASE, aggregated sDA and ASE                                     |
| [30] | Daylight  | KNN, SVM, RF                   | Solar altitude and azimuth, DNI and DHI                                               | Binary glare status for each venetian blind slat angle                            |
| [31] | Daylight  | ANN                            | Geometric, material, orientation, climate and sensor-location                         | Point-wise annual UDI components; Derived UDI and DA spatial distributions        |
| [32] | Daylight  | ANN (MLP)                      | Sensor-obstacle ray distances and window position-direction descriptors               | Per-sensor UDIF, UDIS, UDIA, UDIE percentages                                     |
| [33] | Daylight  | ANN (MLP)                      | Parametric cuboid building layout, heights, spacing, orientation, sun position        | Shading time intervals and neighborhood-scale sunlight-hour heatmaps              |
| [34] | Daylight  | XGB, RFR, DTR with SHAP-xAI    | KF module motor angles and outdoor daylight-climate variables                         | DGP and indoor illuminance at sensor                                              |
| [35] | Daylight  | Multimodal Pix2Pix             | Solid-angle spatial grayscale maps plus material and weather vectors                  | Indoor illuminance distribution cloud maps                                        |
| [36] | Daylight  | ANN, Pix2Pix, XGB              | Numerical and image encodings of room, window and obstructions                        | Point-wise DF values, room DF maps, and standard compliance                       |
| [37] | Daylight  | RTE- and diffusion-based PINNs | Parametric office geometry, window properties, and surface reflectivity               | Predicted daylight autonomy values across interior sensor grid                    |
| [38] | Daylight  | Multimodal GAN-CNN             | Semantic spatial grids and material, climate, and view variables                      | Luminance distribution maps and DGP-based glare performance metrics               |
| [39] | Daylight  | Multimodal GAN with CBAM, GAT  | Floorplan images, room graphs, building attributes, climate conditions                | UDI distribution images for residential buildings                                 |
| [40] | Daylight  | ANN                            | Variables of window geometry, orientation, glass properties and sensor node viewpoint | Annual hourly illuminance and operative temperature                               |
| [41] | Energy    | Metaheuristic-optimized ANN    | Early-design variables of residential buildings                                       | Annual energy consumption of residential buildings                                |
| [42] | Energy    | ResNet-style TCN               | Weather time series and building design parameters                                    | Annual heating and cooling demand and hourly heating demand                       |
| [43] | Energy    | XGB, GB, RF, AdaBoost, LR      | Operational schedules and cooling setpoints                                           | Annual and peak cooling load intensities                                          |
| [44] | Energy    | MLP, RNN, and HGBoost          | Hourly climate variables and precinct-scale geometric and                             | Hourly site energy use profiles for high-rise residential precincts               |

(continued on next page)

Table 1 (continued)

| Ref  | Objective | Methods                        | Inputs                                                                                          | Outputs                                                              |
|------|-----------|--------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| [45] | Energy    | GAT-based GNN                  | spatial-abstract features<br>Graphs of residential blocks with node and edge geometric features | Node-level EUI, TCP and UDI predictions for all buildings            |
| [46] | Energy    | XGB                            | 19 design features including climate factors                                                    | LCCI, DA, cDA, UDI, GA                                               |
| [47] | Radiation | 3D CNN                         | Voxelized 3D building and boundary geometry with ternary occupancy                              | Voxel-level annual solar radiation intensities on building facades   |
| [48] | Radiation | PGML-NN                        | Monthly GHI, ambient temperature, and clearness index at each site                              | Building-scale rooftop PV potential (planet-scale)                   |
| [49] | Radiation | PointNet-based CNN, T-Net, MLP | Weighted point clouds with normals, WWR, and orientation angles                                 | Annual block-level PV potential                                      |
| [50] | Radiation | MLGA-GNN                       | Graph-based building and urban morphology features with geo-spatial edges                       | Layer-wise facade solar radiation intensity and PV potential metrics |
| [51] | Radiation | SLP, MLP, 1D/2D-CNN, RFR, SVR  | Parametric block/building variables, solar/sky data                                             | Building-scale solar potential (roof + facades)                      |
| [52] | Thermal   | PINN-based 2R2C-RC NN          | Indoor and outdoor temperature, HVAC actions, 2R2C RC parameters                                | Room air temperature, thermal mass temperature, cooling load demand  |

driven surrogates have been proposed at building, district and planetary scales. Han et al. (2022) [47] adopt 3D CNNs to predict voxel-level annual solar radiation intensity on building facades from voxelized building and boundary geometry, effectively replacing Radiance for facade radiation estimation. Tian et al. (2025) [51] combine parametric block/building variables with solar/sky data and train multiple ML models (SLP, CNN, RFR, etc.) to estimate building-scale solar potential for roofs and facades, reducing evaluation time from hundreds of seconds to under 2 s. Geng et al. (2025) [49] use PointNet-like networks on weighted point clouds with normals, window-to-wall ratio (WWR), and orientation angles to predict block-level BIPV energy potential for 400 m × 400 m urban units. Li et al. (2025) [50] introduce an MLGA-GNN framework that encodes buildings and urban morphology as graphs and predicts layer-wise facade solar radiation intensity and PV potential, enabling vertical stratification of BIPV deployment and economic analysis.

Technically, these studies show that, when complete or highly structured 3D representations are available, such as city or building models, dense point clouds or urban graphs, learning-based surrogates can substantially reduce the cost of large-scale radiation and BIPV assessment. From the standpoint of conceptual design, however, they remain misaligned in three respects. First, their inputs typically rely on fully specified models or dense scans that must be generated through file conversions, meshing, registration or other preprocessing steps, and therefore cannot be obtained directly from the tools used in early-stage workflows. Second, their outputs are often aggregated indicators at the building, layer, or district level, or volumetric fields in 3D coordinates, which are not readily interpretable as view-aligned facade feedback and thus cannot be used as immediate, actionable guidance during iteration.

Third, even when inference is fast, the overall workflow is usually split across separate modeling, export, analysis, and visualization environments, creating a fragmented loop that prevents performance feedback from being delivered seamlessly within the interactive design software where decisions are actually made.

In short, despite their accuracy and speed, existing solar radiation surrogates still lack a view-aligned, facade-level framework that can take an architect's current design view as input and directly return the annual solar radiation distribution over the visible exterior surfaces.

#### 2.4. Multi-climate surrogates

Climate environments and solar potential vary greatly from city to city, and to ensure the broad applicability and reliability of design tools, cross-climate surrogates have therefore received growing attention [46]. Westermann et al. (2020) [42] use a ResNet-style temporal convolutional network (TCN) to learn heating and cooling demand directly from raw weather time series and building parameters, showing strong generalization across weather stations. Kristiansen et al. (2022) [40] employ ANNs to predict hourly illuminance and operative temperature at multiple sensor nodes, highlighting the role of shortwave radiation in thermal comfort. Govindarajan et al. (2025) [44] generate synthetic residential precincts in Singapore and train MLP and gradient-boosted ensembles to predict site energy use time series from hourly climate variables and precinct-scale geometric features.

Collectively, these studies show that incorporating weather time series or compressed climate indicators (e.g., degree-days, monthly aggregates) as part of the input can yield unified energy surrogates that transfer across multiple locations. In this study, we extract compact climate descriptors from EPW files and fuse them with camera pose to train a climate-conditioned surrogate model under 20 representative climate conditions.

#### 2.5. Research gaps and positioning

Overall, the main gaps that our work seeks to address can be summarized as follows: (i) Most sketch-based methods use sketches primarily for geometry reconstruction, while environmental assessment is still deferred to downstream simulation rather than predicting climate-driven façade fields directly from design views. (ii) Existing image/grid-based surrogates have largely focused on indoor lighting and often assume clean, structured inputs, leaving façade-level annual solar-radiation distributions on building exteriors underexplored. (iii) Prior solar-radiation and BIPV surrogates typically require fully specified 3D models or dense representations and produce outputs that are not view-aligned, which mismatches early-stage, view-aligned iteration. (iv) While multi-climate surrogates show the value of climate descriptors for cross-location generalization, a unified framework that fuses view information with climate conditioning to predict façade-level annual solar-radiation distributions in a view-aligned manner remains missing.

To fill these gaps, we propose a view-aligned surrogate that takes a single design view as input and outputs a directly interpretable, view-aligned façade radiation map, enabling rapid screening and comparative iteration.

### 3. Methods

#### 3.1. Dataset and preprocessing

Existing public datasets lack sufficiently large-scale cross-climate building radiation data. This study constructs a large-scale, multi-view, multi-climate dataset based on 1000 real buildings in Manhattan. Each building is rendered from 24 viewpoints to obtain sketch images, depth maps and Radiance-based annual radiation map. The same geometry is then re-simulated under 20 representative climates, yielding a dataset of 20 cities × 1000 buildings × 24 views. Climate descriptors are fused



with camera pose at the network bottleneck, enabling the surrogate to adapt its predictions across climates and buildings.

Fig. 2 illustrates the workflow for generating the dataset in this study. The raw building geometry used in this study comes from the Manhattan-1000 dataset [26], a curated collection of real building models from the RealCity3D repository [60], originally assembled for sketch-based mesh reconstruction. The corpus spans diverse urban typologies and is stratified into small, medium and large complexity classes to support controlled evaluation. To ensure the geometry is suitable for Radiance simulation, we clean these meshes by unifying surface normals and removing duplicate, non-manifold, and zero-area faces, producing watertight, simulation-ready models for Radiance-based evaluation.

Each building is rendered from 24 adaptively chosen views. Camera distance and field of view are scaled according to building height and footprint, ensuring that most views capture the full outline while preserving sufficient depth variation and self-occlusion. All building and camera coordinate systems are aligned in Rhino to ensure consistent azimuth semantics across the entire dataset. Raw camera parameters are stored as spherical coordinate parameters (azimuth, elevation, roll, scaling factor and focal length) and normalized to a compact camera vector during preprocessing, serving as part of the conditional input. For each building and its 24 views, we generated two  $512 \times 512$  pixel sketches: a standard sketch and a hand-drawn sketch with line jitter and stroke variations. The depth map (Z buffer) was exported using Rhino's ShowZBuffer module.

To capture cross-climate characteristics, we selected 20 representative cities: Auckland, Beijing, Brisbane, Buenos Aires, Cairo, Dubai, Hong Kong, London, Los Angeles, Melbourne, Moscow, Mumbai, Nairobi, New York, Ottawa, Paris, Santiago, Seoul, Singapore and Ushuaia. For each city, we extracted its latitude, longitude, monthly mean hourly DNI and DHI. Among them, DNI and DHI mainly determine the intensity and contrast of the radiation field and are widely used in the assessment of solar radiation and building surface potential [61]. These 26 scalar values were concatenated into a low-dimensional climate descriptor  $c \in \mathbb{R}^{26}$  and normalized across cities.

Fig. 3 shows that DNI and DHI exhibit significant heterogeneity across different cities and months, rather than simple synchronous changes. On the one hand, cities in the Northern and Southern Hemispheres show opposite phase characteristics in seasonal variations. On the other hand, high-latitude cities have larger seasonal amplitudes, while some cities with monsoon or cloudy climates show more

significant variations in specific months. In contrast, other common meteorological variables, such as temperature and humidity, mainly affect facade irradiance indirectly through atmospheric scattering and cloud cover, and therefore are not included in the minimum descriptor of this study.

We developed a parameterized workflow in Grasshopper to generate the dataset (Fig. 4). Annual solar radiation maps are simulated in ClimateStudio using Radiance with high-accuracy settings (-ab 7, -ad 4096, -aa 0.0, -ar 64, -as 1024, -lw 0.01). The ground is modeled as asphalt with a fixed albedo to approximate typical urban surface conditions. This study focuses on the volumetric relationships and self-occlusion of target geometry, while the contextual occlusion problem in dense urban blocks will be analyzed and compared separately in Section 5. We apply a fixed pseudo-color mapping to encode radiation values as 8-bit RGB values, outputting the result as a  $512 \times 512$  resolution PNG image. Each pixel's 3-channel RGB vector corresponds to a normalized radiation value, recording the underlying physical units and color scale so that these values can be reconstructed if necessary (Table S1).

Solar radiation simulation constitutes the primary computational bottleneck. With GPU acceleration, each case requires approximately 15–90 s, depending on geometric complexity. For each building-city pair, the simulation is performed once and reused to render 24 viewpoints. All experiments were conducted on a workstation equipped with an AMD Ryzen 9 9950 × 3D CPU and a single NVIDIA GeForce RTX 5080 GPU. Under this setup, generating the full set of 480,000 radiation images requires approximately 300 h.

All inputs and outputs are stored as compressed NPZ archives for efficient loading during training and evaluation, and are consistently indexed by city, building, and view identifiers. To prevent geometry leakage, data are split at the building level. Specifically, the 1000 buildings are stratified into three geometry-complexity categories (small, medium and large), followed by a 70/20/10 train/validation/test split that preserves comparable class distributions across subsets. This building-wise protocol ensures that evaluation reflects generalization to unseen geometries, rather than memorization of recurring façade patterns observed under different viewpoints or climate conditions.

### 3.2. Network architecture

Each input sample consists of a sketch  $S$  and a Z-buffer  $Z$ , concatenated into a tensor  $[S, Z] \in \mathbb{R}^{512 \times 512 \times 2}$ . The camera pose  $p$  and climate descriptor  $c$  are encoded separately and injected as a condition. The

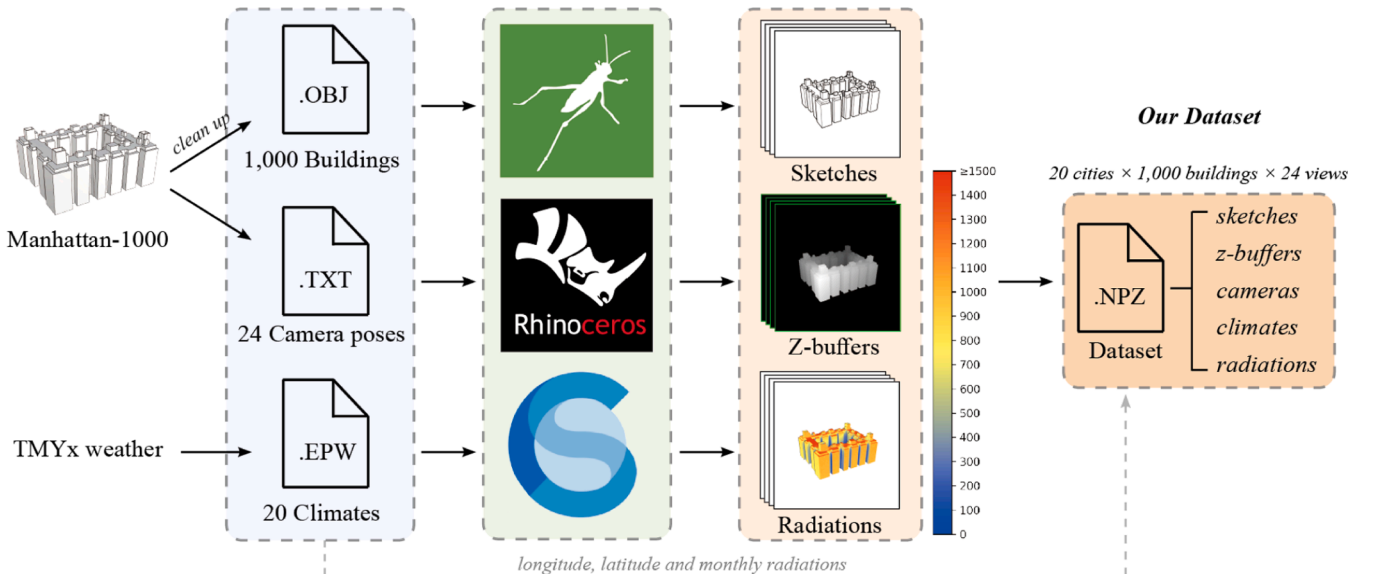


Fig. 2. The source and structure of the dataset used in this study.

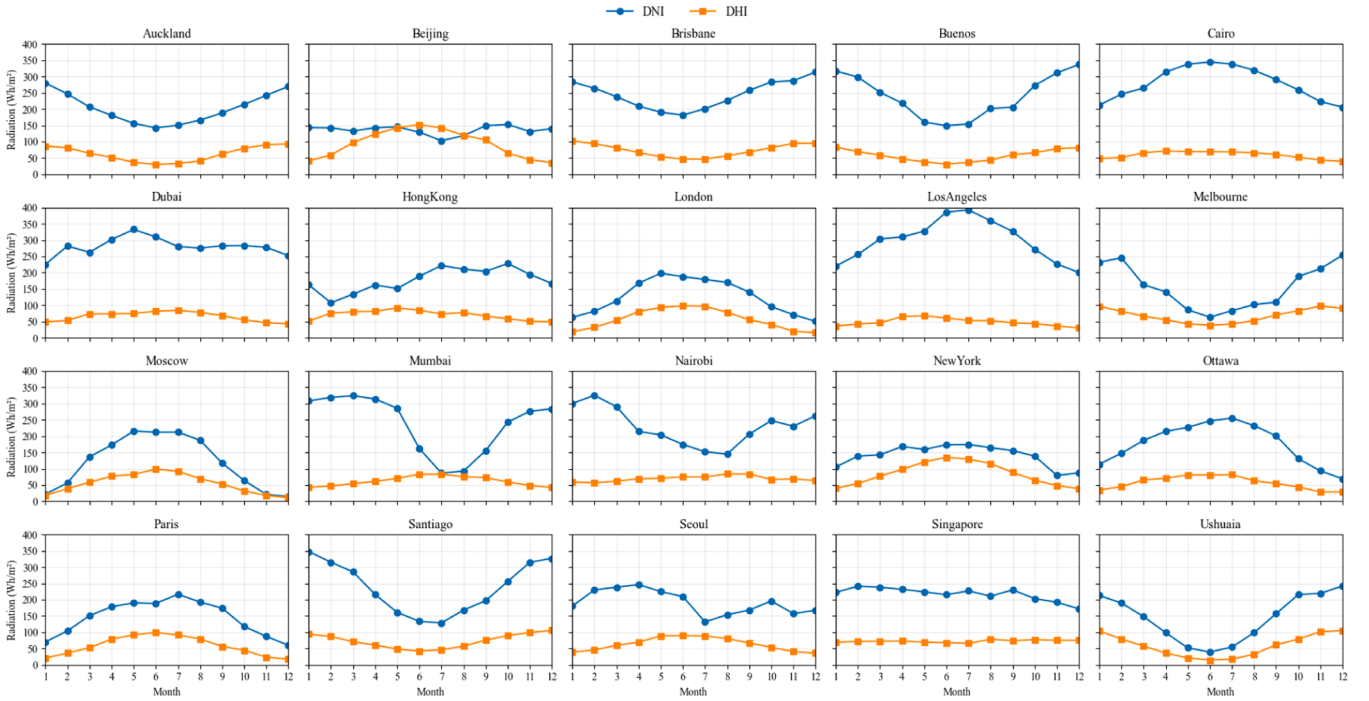


Fig. 3. Monthly profiles of mean hourly direct normal irradiance (DNI) and diffuse horizontal irradiance (DHI) across 20 cities.

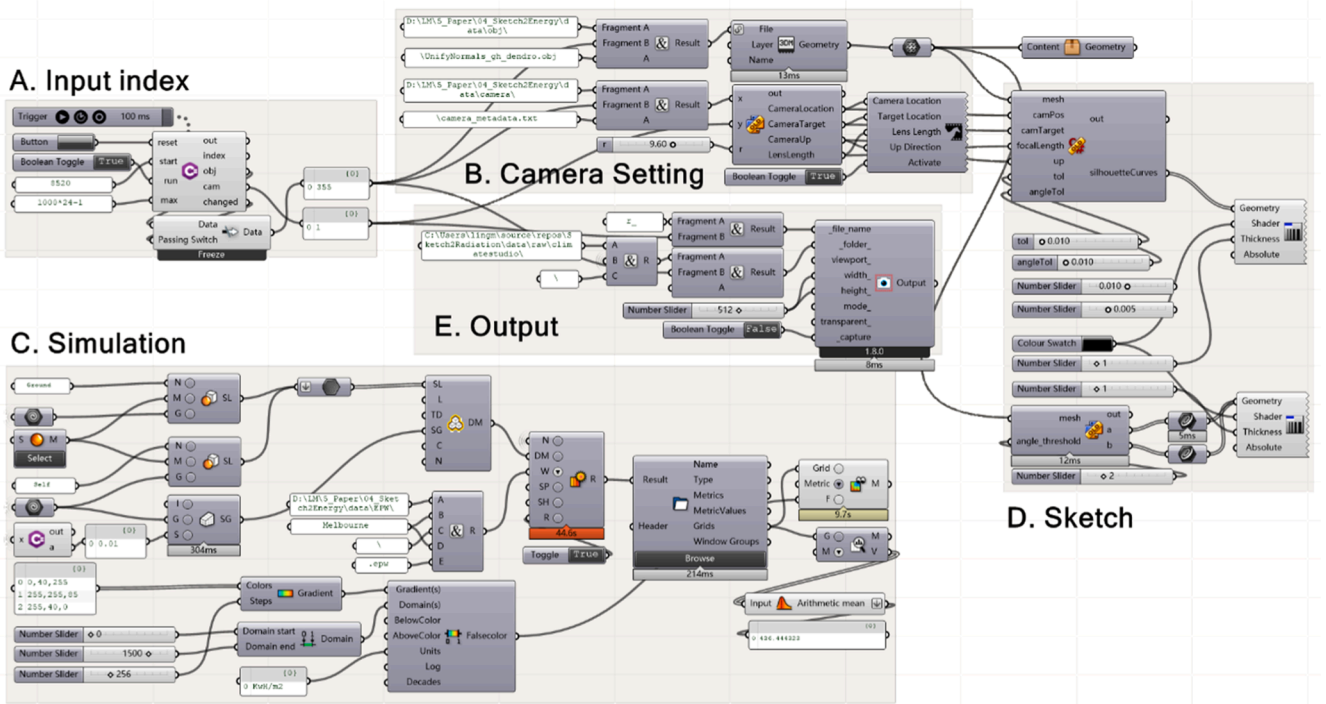


Fig. 4. Parameterized workflow for generating datasets in Grasshopper.

network predicts a 3-channel radiation field  $I \in \mathbb{R}^{512 \times 512 \times 3}$  and uses a unified gradient color legend for visualization. Our framework follows a Pix2Pix-style cGAN [53] with a UNet-based generator and a PatchGAN discriminator (Fig. 5).

**Generator (UNet).** An encoder-decoder with six down-sampling stages compresses the  $512 \times 512 \times 2$  input to an  $8 \times 8$  latent tensor. The pose vector  $p$  is processed by a small MLP, reshaped to the bottleneck size, and added element-wise to the latent features to

provide global, view-aware context without disturbing spatial encoding. The decoder mirrors the encoder with six up-sampling stages (transpose-convs or upsample+conv) and skip connections linking symmetric layers, preserving fine-grained geometric cues from the sketch and depth inputs. The final layer outputs a non-negative 3-channel radiation map, encouraging physical plausibility. **Discriminator (PatchGAN).** A patch-level realism critic receives the concatenation  $[S, Z, \tilde{I}]$  or  $[S, Z, I]$ , where  $\tilde{I}$  is the generated (fake) one and  $I$  is the simulated (real) one. Patch-wise scoring enforces spatial

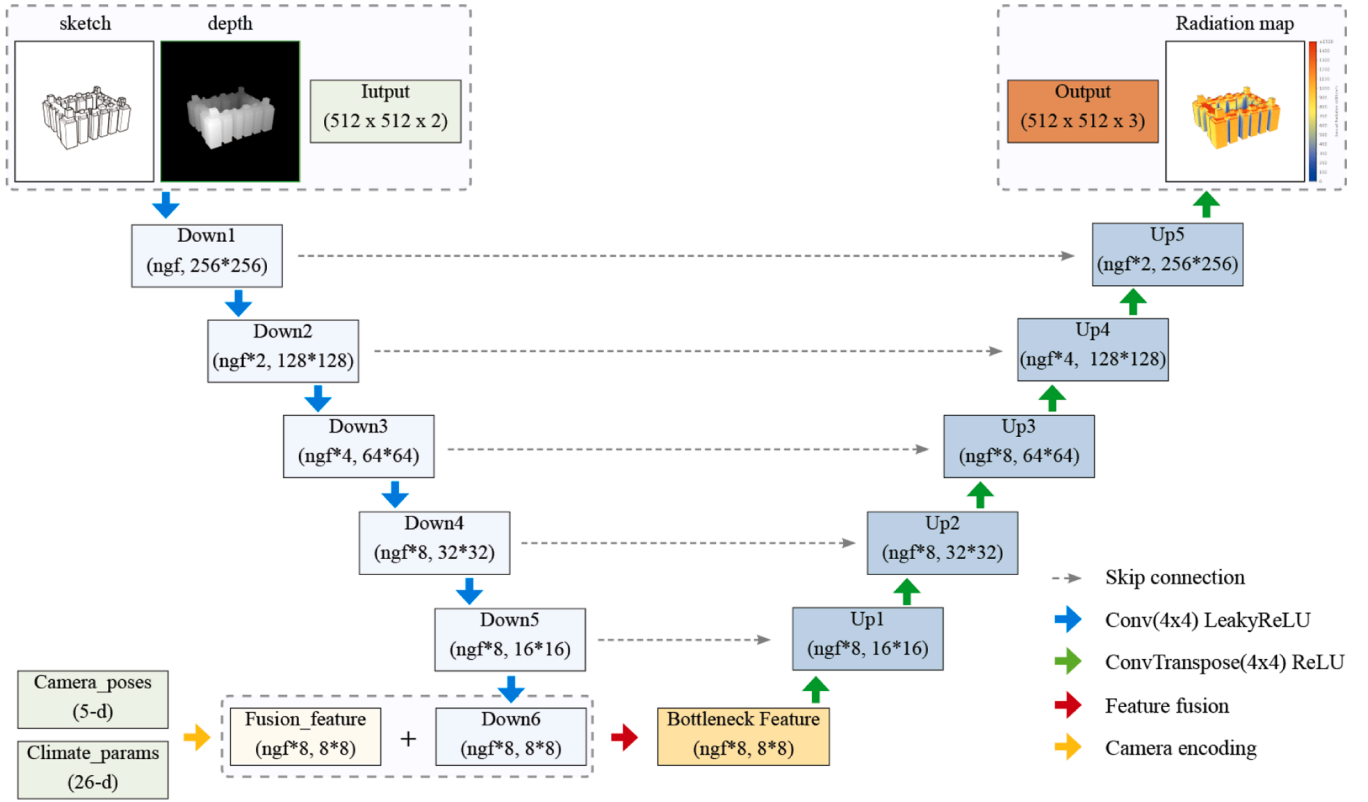


Fig. 5. UNet-based generator architecture with camera- and climate-conditioned bottleneck.

consistency along edges and depth discontinuities while avoiding global mode collapse.

This design achieves a balance between inference speed and geometric fidelity: skip connections preserve high-frequency details of edges and depth gradients, explicit pose conditions stabilize view sensitivity, and PatchGAN promotes locally consistent radiative textures.

### 3.3. Training and implementation details

We train the model using a combination of pixel-wise L1 reconstruction loss and conditional adversarial loss, with perceptual loss as an option. **Adversarial loss** ( $\mathcal{L}_{GAN}$ ) ensures training stability via the least-squares GAN objective. **Pixel-wise reconstruction loss** ( $\mathcal{L}_{L1}$ ) enforces pixel-level numerical consistency, especially in low-frequency components. **Perceptual loss** ( $\mathcal{L}_p$ ) computed from the relu3\_3 layer of a pretrained VGG19 to enhance structural sharpness and perceptual interpretability. The final objective combines the three terms:

$$G^* = \arg \min_G \max_D [\mathcal{L}_{GAN}(G, D) + \lambda_{L1} \cdot \mathcal{L}_{L1} + \lambda_p \cdot \mathcal{L}_p] \quad (1)$$

We set  $\lambda_{L1} = 100$  and  $\lambda_p = 0.1$ . Both generator and discriminator are optimized with Adam, using an initial learning rate of  $5 \times 10^{-5}$ . *CosineAnnealingWarmRestarts* is applied to both optimizers, with  $T_0 = 20$ ,  $T_{mult} = 1$ , and  $eta_{min} = 5 \times 10^{-7}$ . Training is conducted with automatic mixed precision (AMP) to accelerate computation and enhance numerical stability. Early stopping with a patience of 10 epochs is employed to mitigate overfitting. Data are loaded using PyTorch DataLoaders with a batch size of 16. On the test set, the network forward pass takes  $9.1 \times 10^{-5}$  s per sample on average (excluding preprocessing and I/O), enabling rapid model-side feedback for early-stage design iteration. The following sections examine the Grasshopper integration and end-to-end latency optimizations.

## 4. Experiments and results

### 4.1. Experimental setup and evaluation metrics

The proposed method is evaluated along three complementary dimensions: physical consistency, perceptual similarity and design usefulness. Physical consistency is assessed by comparing the predicted radiation maps against Radiance simulations at the pixel level, while perceptual similarity is examined through a two-alternative forced-choice (2AFC) study involving participants with architectural design experience. Design usefulness is evaluated using task-relevant indicators that reflect the model's ability to support early-stage BIPV decision-making.

Experiments are conducted under both single-city and multi-city settings. In the single-city setup, separate models are trained using one EPW file per city to establish baseline performance. In the multi-city setup, samples from 20 cities are combined, and climate descriptors are incorporated to enable climate-conditioned prediction. Model performance is analyzed at both pixel and building scales.

At the pixel level, prediction accuracy is quantified using MSE, MAE, SSIM, FSIM and PSNR [62–65], capturing both numerical fidelity and structural similarity between predicted and simulated radiation maps. At the building level, two design-oriented indicators are introduced to complement pixel-wise evaluation, including the total error and distributed error of annual radiation. These building level indicators are intended to reflect performance in terms that are directly relevant to BIPV planning and early-stage design decision making. Their detailed definitions and interpretations are provided in the following subsection.

### 4.2. Single-city surrogate model

In this experiment, we select three representative cities for testing and train surrogate models for each city. As shown in Table 2, surrogates achieve high pixel-level accuracy and strong structural consistency

**Table 2**

Performance of the single-city surrogate model across building scales in New York, Hong Kong and Melbourne.

| Cities    | Test set       | MSE              | MAE              | SSIM             | FSIM             | PSNR         |
|-----------|----------------|------------------|------------------|------------------|------------------|--------------|
| New York  | Small (792)    | 0.0002799        | 0.0023779        | 0.9939632        | 0.9969526        | 41.21        |
|           | Medium (792)   | 0.0002168        | 0.0016500        | 0.9924123        | 0.9959022        | 40.38        |
|           | Large (816)    | 0.0004034        | 0.0032462        | 0.9863424        | 0.993696         | 37.59        |
|           | Overall (2400) | <b>0.0002998</b> | <b>0.0024242</b> | <b>0.9909356</b> | <b>0.9955309</b> | <b>39.74</b> |
| Hong Kong | Small (792)    | 0.0002497        | 0.0017618        | 0.9955196        | 0.9973836        | 42.79        |
|           | Medium (792)   | 0.0001780        | 0.0016799        | 0.9943475        | 0.9966564        | 41.94        |
|           | Large (816)    | 0.0003976        | 0.0027535        | 0.988328         | 0.9943816        | 38.68        |
|           | Overall (2400) | <b>0.0002749</b> | <b>0.0020621</b> | <b>0.9927588</b> | <b>0.9961526</b> | <b>41.15</b> |
| Melbourne | Small (792)    | 0.0002040        | 0.0019905        | 0.9947431        | 0.9974332        | 42.05        |
|           | Medium (792)   | 0.0001604        | 0.0019416        | 0.9934267        | 0.9967031        | 41.19        |
|           | Large (816)    | 0.0003548        | 0.0029720        | 0.9868107        | 0.9944481        | 38.25        |
|           | Overall (2400) | <b>0.0002394</b> | <b>0.0022983</b> | <b>0.9916901</b> | <b>0.9962068</b> | <b>40.51</b> |

across all building scales. While large-scale buildings show larger pixel errors compared to small and medium-sized buildings, their perceptual metrics (SSIM, FSIM, PSNR) remain consistently high, indicating that structural patterns and contrast are well preserved. These results confirm the model's robustness under both geometric and climatic variation.

The qualitative comparison in Fig. 6 shows that the facade radiation gradient caused by orientation and self-shading is faithfully reproduced across different scales and climatic conditions. Features such as facade details and roof setbacks are also consistently captured, supporting the interpretability of the predicted outputs.

#### 4.3. Multi-city surrogate model

In the multi-city experiment, we pool samples from 20 cities and condition the network on a compact climate descriptor to learn how the same geometry responds to different climates. On the test set (Table 3), the multi-city model achieves MSE, MAE, SSIM, and FSIM that are comparable to the single-city baselines, suggesting that explicit climate conditioning preserves accuracy in facade-scale radiation prediction.

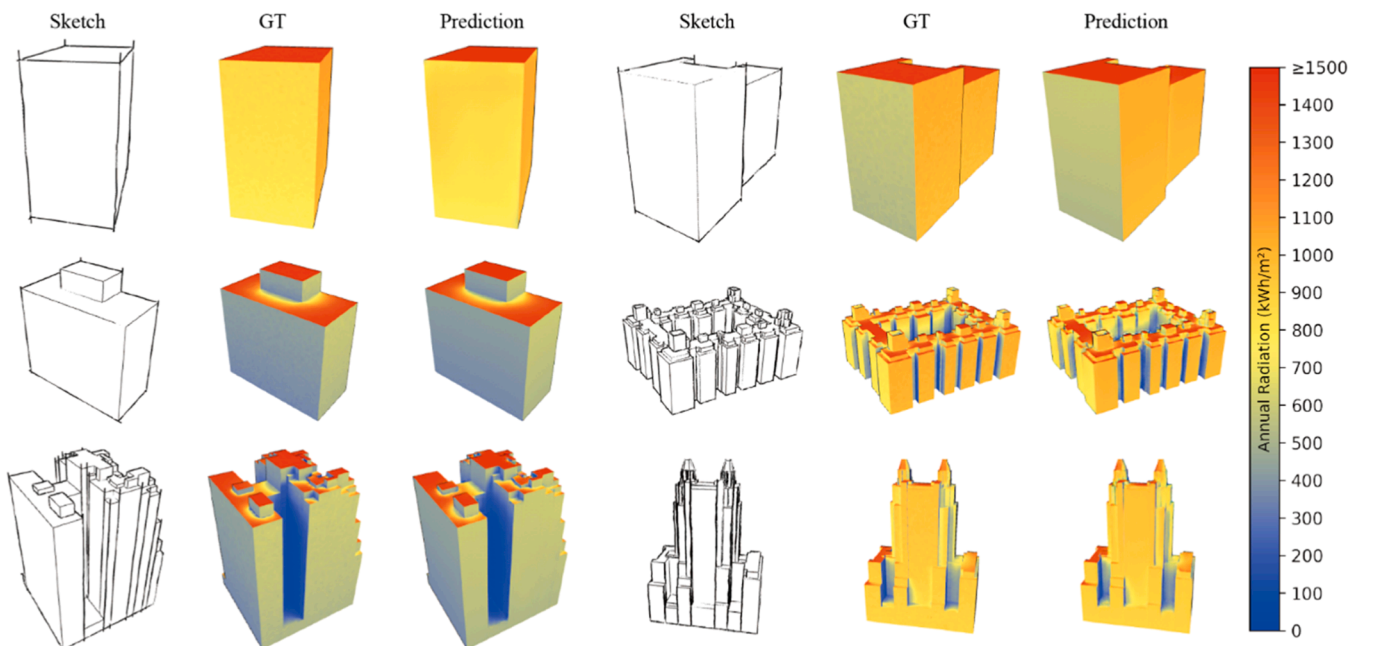
Fig. 7 further shows that, for the same building across different cities, the model reproduces expected shifts in radiation patterns—such as differences between low- and high-latitude locations and the reversal of solar-path orientation between the Northern and Southern Hemispheres. This indicates that the model leverages the coupling between the climate

**Table 3**

Pixel-level performance (MSE, MAE, SSIM, FSIM, PSNR) of the multi-city surrogate model on the test set.

| Cities       | MSE             | MAE             | SSIM            | FSIM            | PSNR         |
|--------------|-----------------|-----------------|-----------------|-----------------|--------------|
| Auckland     | 0.000220        | 0.002273        | 0.990977        | 0.995698        | 39.56        |
| Beijing      | 0.000210        | 0.002227        | 0.990155        | 0.995477        | 39.43        |
| Brisbane     | 0.000249        | 0.002358        | 0.990786        | 0.995822        | 39.14        |
| Buenos Aires | 0.000295        | 0.002453        | 0.989944        | 0.995391        | 38.38        |
| Cairo        | 0.000241        | 0.002096        | 0.991283        | 0.996180        | 39.40        |
| Dubai        | 0.000242        | 0.002262        | 0.990533        | 0.995874        | 39.17        |
| Hong Kong    | 0.000258        | 0.002428        | 0.989535        | 0.995694        | 39.00        |
| London       | 0.000192        | 0.002280        | 0.990126        | 0.995257        | 39.80        |
| Los Angeles  | 0.000252        | 0.002139        | 0.991292        | 0.996062        | 39.53        |
| Melbourne    | 0.000228        | 0.002539        | 0.989531        | 0.995651        | 39.37        |
| Moscow       | 0.000218        | 0.002354        | 0.989583        | 0.995211        | 39.47        |
| Mumbai       | 0.000193        | 0.001925        | 0.992271        | 0.996522        | 40.23        |
| Nairobi      | 0.000263        | 0.002512        | 0.989254        | 0.994740        | 38.75        |
| New York     | 0.000274        | 0.003227        | 0.988181        | 0.994918        | 37.53        |
| Ottawa       | 0.000212        | 0.002281        | 0.990752        | 0.995810        | 39.56        |
| Paris        | 0.000219        | 0.002310        | 0.989489        | 0.995094        | 39.36        |
| Santiago     | 0.000226        | 0.002321        | 0.991155        | 0.995877        | 39.35        |
| Seoul        | 0.000203        | 0.002129        | 0.991379        | 0.996108        | 39.58        |
| Singapore    | 0.000245        | 0.002506        | 0.990097        | 0.994993        | 38.90        |
| Ushuaia      | 0.000281        | 0.002749        | 0.988216        | 0.994436        | 38.65        |
| Overall      | <b>0.000236</b> | <b>0.002371</b> | <b>0.990219</b> | <b>0.995537</b> | <b>39.20</b> |

descriptor and the view-aligned geometric cues (depth and perspective).



**Fig. 6.** Visual comparison of the surrogate model's predictions and Ladybug results.



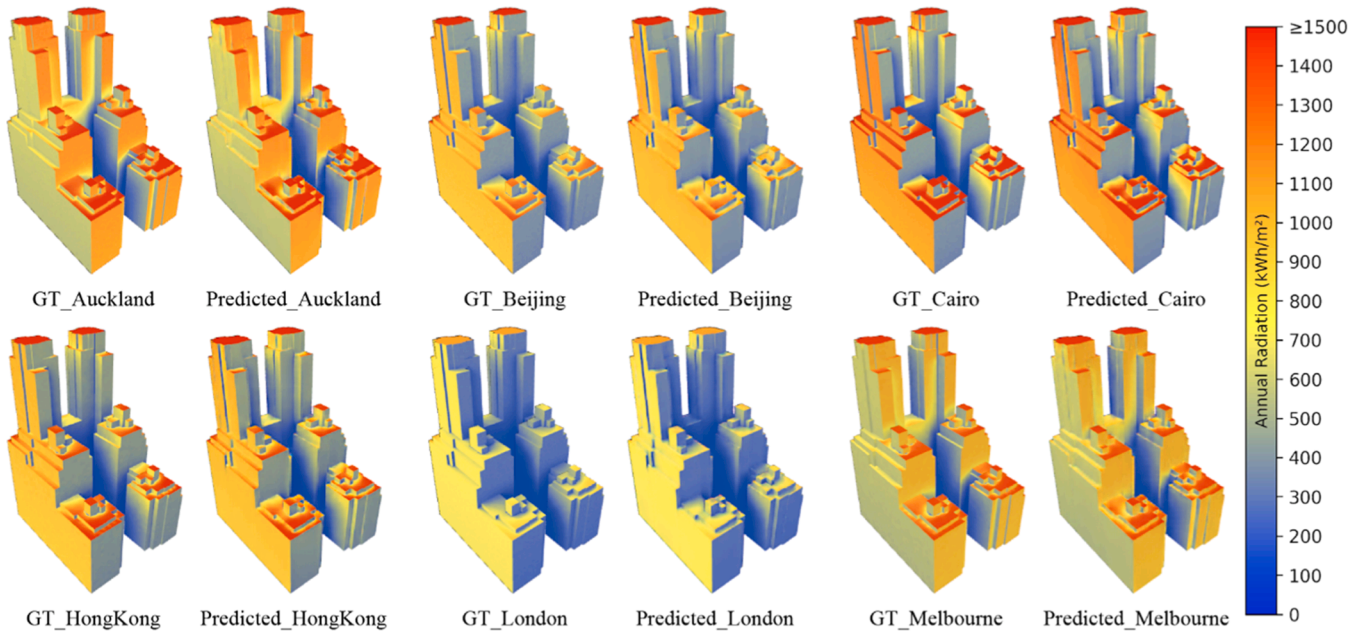


Fig. 7. Comparison of ground truth and predicted result of different cities based on the same building geometry and view.

To evaluate the robustness of the surrogate model under varying climatic conditions, a lightweight sensitivity analysis was conducted by grouping the 20 cities into three latitude bands: low latitude ( $|\text{latitude}| < 23.5^\circ$ ), mid latitude ( $23.5^\circ - 45^\circ$ ), and high latitude ( $|\text{latitude}| > 45^\circ$ ). Group-wise averaged prediction metrics are summarized in Table 4.

The results indicate generally stable performance across latitude bands, with only minor differences in averaged metrics. The mid-latitude group shows the most consistent overall performance, achieving the highest average SSIM (0.990474) and FSIM (0.995733). The high-latitude group attains the highest average PSNR (39.37 dB), although its SSIM and FSIM are slightly lower than those of the other groups. The low-latitude group performs comparably to the mid-latitude group across all metrics. Overall, no clear or monotonic trend of performance degradation with latitude is observed, suggesting that latitude alone is not a dominant factor influencing model accuracy.

This experiment is intended to validate feasibility under known climatic conditions, not to claim full zero-shot generalization to unseen climates. For cities outside the 20-EPW set, the model would still require retraining or lightweight fine-tuning with a small amount of local data. As future work, we will further improve interpretability by explicitly incorporating additional irradiance descriptors and conducting component-wise analyses that separate direct and diffuse contributions in simulation.

#### 4.4. Sensitivity analysis of orientation

We evaluate the model's sensitivity to orientation through two complementary tests: (i) varying the camera azimuth to change the viewing direction, and (ii) rotating the building geometry to alter its physical orientation.

In the first test (Fig. 8a), we sweep the camera azimuth across the full  $0-360^\circ$  range in  $30^\circ$  increments. This keeps a comparable facade view in

image space while changing its solar orientation. The predicted radiation maps evolve smoothly and consistently: the same surfaces increase or decrease as expected with changing solar exposure. This indicates that the model effectively uses orientation information encoded in the camera pose.

In the second test (Fig. 8b), we fix the camera pose and rotate the building in  $30^\circ$  increments. Although different facade portions come into view, cases that share the same absolute azimuth (and thus similar lighting conditions) yield highly consistent global shading and radiation patterns, with differences mainly confined to local geometric details. This suggests that the model has learned a stable mapping from solar orientation to scene-level annual radiation, rather than relying on low-level appearance cues.

Together, these experiments show that the framework is robust to both view-driven and geometry-driven changes in orientation. By combining sketch cues with camera metadata, the model supports rapid, orientation-aware solar reasoning in early-stage massing studies, when detailed 3D models are not yet available.

#### 4.5. Design-oriented PV metrics

In practice, designers typically care less about pixel-wise deviations and more about decision-relevant quantities, namely the building's total annual solar energy and the share of facade area that falls into PV-suitable radiation levels. Accordingly, two design-oriented metrics are introduced: (i) total error, which captures the deviation in building-scale annual solar potential, and (ii) distributed error, which characterizes how errors vary across radiation intensity ranges.

To exclude background pixels, a binary visible-surface mask  $M \in \{0, 1\}^{H \times W}$  is constructed so that all computations are restricted to building surfaces. The total radiation error for a given (city, building, view) sample is then defined as:

Table 4  
Pixel-level performance of the surrogate model across latitude-based climate zones.

| Groups        | Num | Mean MSE | Mean MAE | Mean SSIM | Mean FSIM | Mean PSNR |
|---------------|-----|----------|----------|-----------|-----------|-----------|
| low-latitude  | 4   | 0.000240 | 0.002343 | 0.990289  | 0.995487  | 39.22     |
| mid-latitude  | 11  | 0.000240 | 0.002366 | 0.990474  | 0.995733  | 39.13     |
| high-latitude | 5   | 0.000224 | 0.002395 | 0.989633  | 0.995162  | 39.37     |

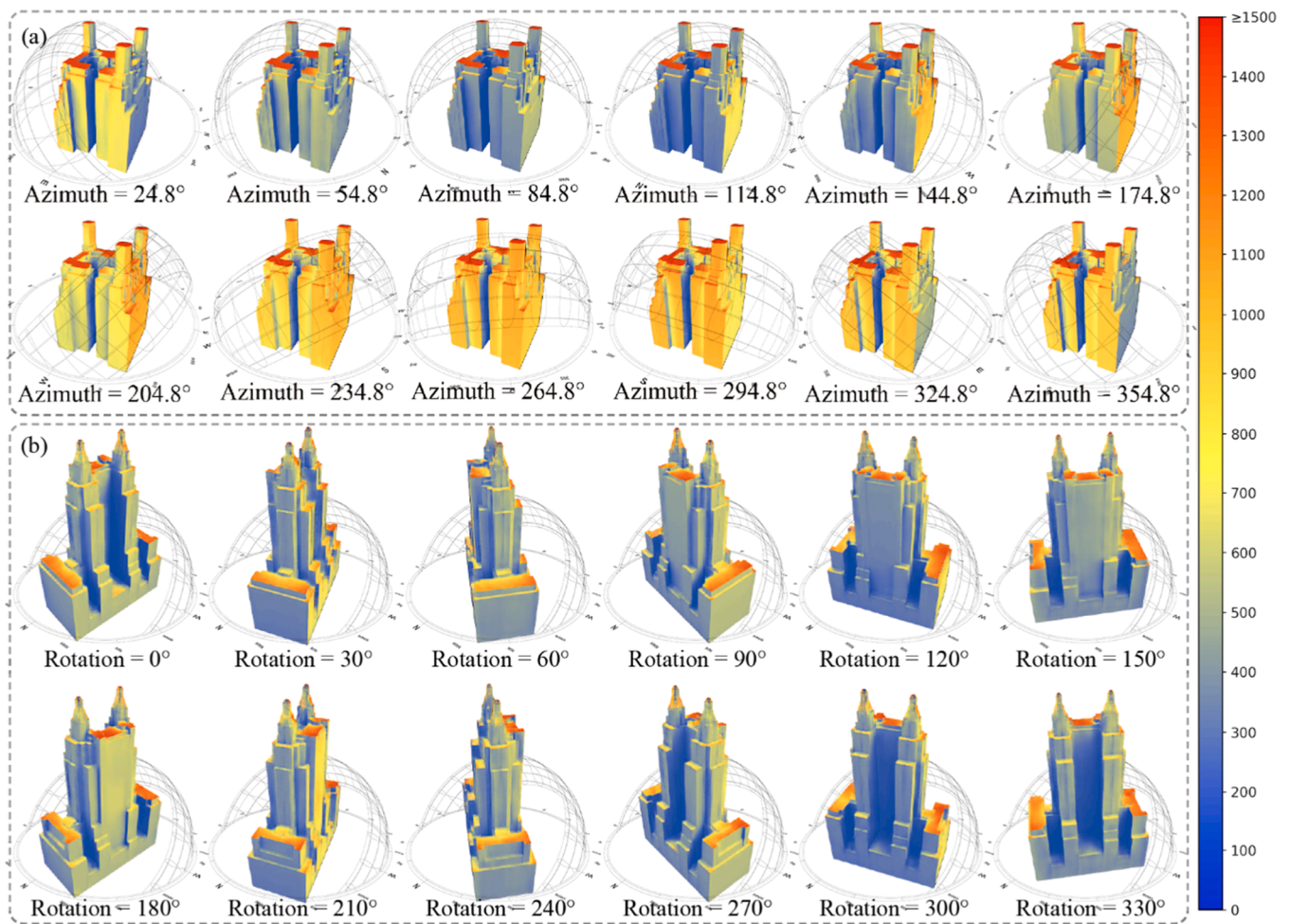


Fig. 8. Sensitivity experiments of building orientation. (a) Varying camera-pose azimuth via camera rotation with fixed geometry; (b) Fixed camera pose with rotating geometry under constant solar azimuth.

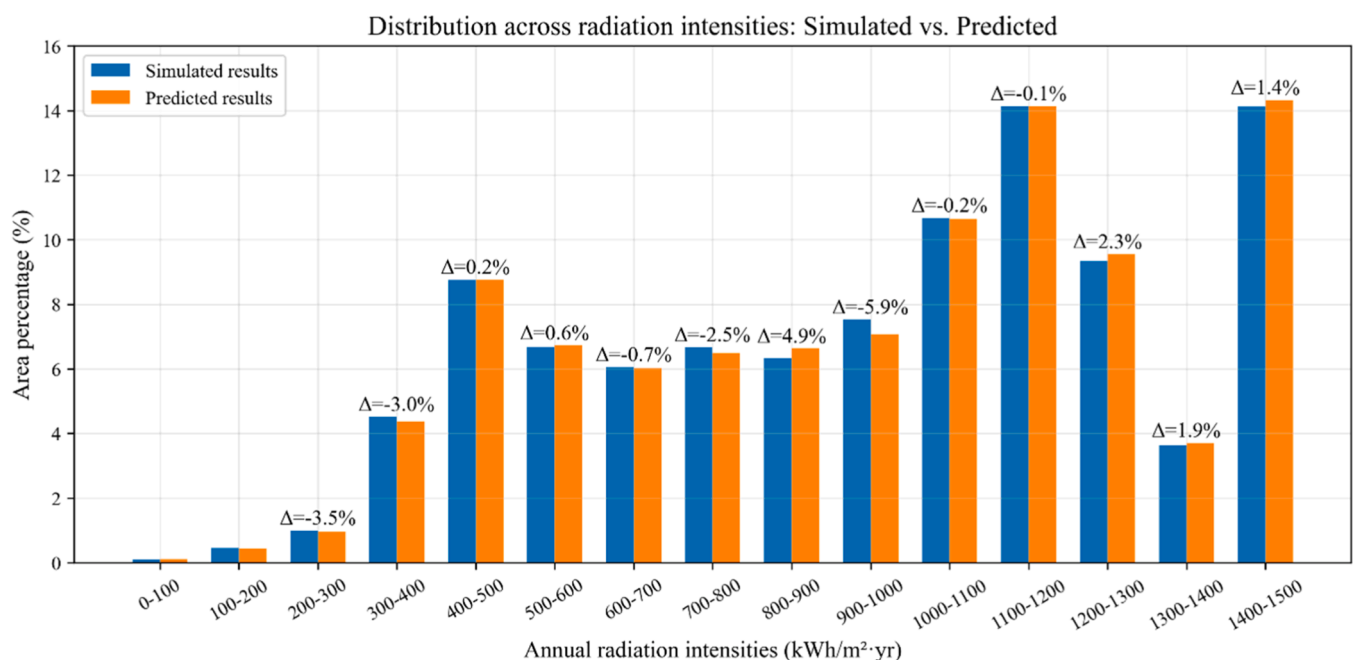


Fig. 9. Distribution across different radiation intensities: Simulated result vs. Predicted result, where  $\Delta$  represents the relative error.

$$e_{\text{total}} = \left| \frac{\sum_{x \in M} y_{\text{pred}}(x) - \sum_{x \in M} y_{\text{gt}}(x)}{\sum_{x \in M} y_{\text{gt}}(x)} \right| \quad (2)$$

Here,  $y_{\text{pred}}(x)$  and  $y_{\text{gt}}(x)$  denote the predicted and ground-truth annual solar radiation at pixel  $x$ , respectively. This metric directly reflects the relative error in estimating building-scale annual solar availability, and therefore corresponds to uncertainty in downstream quantities such as annual PV yield or feasible installed capacity. Across the test set, the total error remains around 0.7%–0.8%, which translates to a deviation of only 7–8 kWh per 1000 kWh of annual total radiation. This magnitude is well below the inherent variability typically associated with long-term meteorological inputs, indicating that Sketch2Radiation provides a reliable basis for comparing alternative massing and facade options during conceptual design.

Beyond the total solar radiation potential, designers also need to know the radiation-level distribution that determines which surface regions are practically usable for BIPV. In practice, PV suitability thresholds vary widely by application and region, with values commonly reported from roughly 440 to 1000 kWh/m<sup>2</sup>·yr [66–68]. Some studies further differentiate thresholds for facades (around 800 kWh/m<sup>2</sup>·yr) and roofs (around 1000 kWh/m<sup>2</sup>·yr) [69,70].

To better assess this aspect, we discretized the annual radiation range into 15 equal intervals. Fig. 9 reports the radiation histogram of area distribution. The relative deviations are concentrated in the medium radiation levels, particularly 700–1000 kWh/m<sup>2</sup>·yr, yet the overall

distribution trend remains consistent across intensities. This indicates that the model robustly captures PV-suitable area fractions over the full radiation spectrum, providing a finer-grained and more actionable basis for facade-level PV siting decisions.

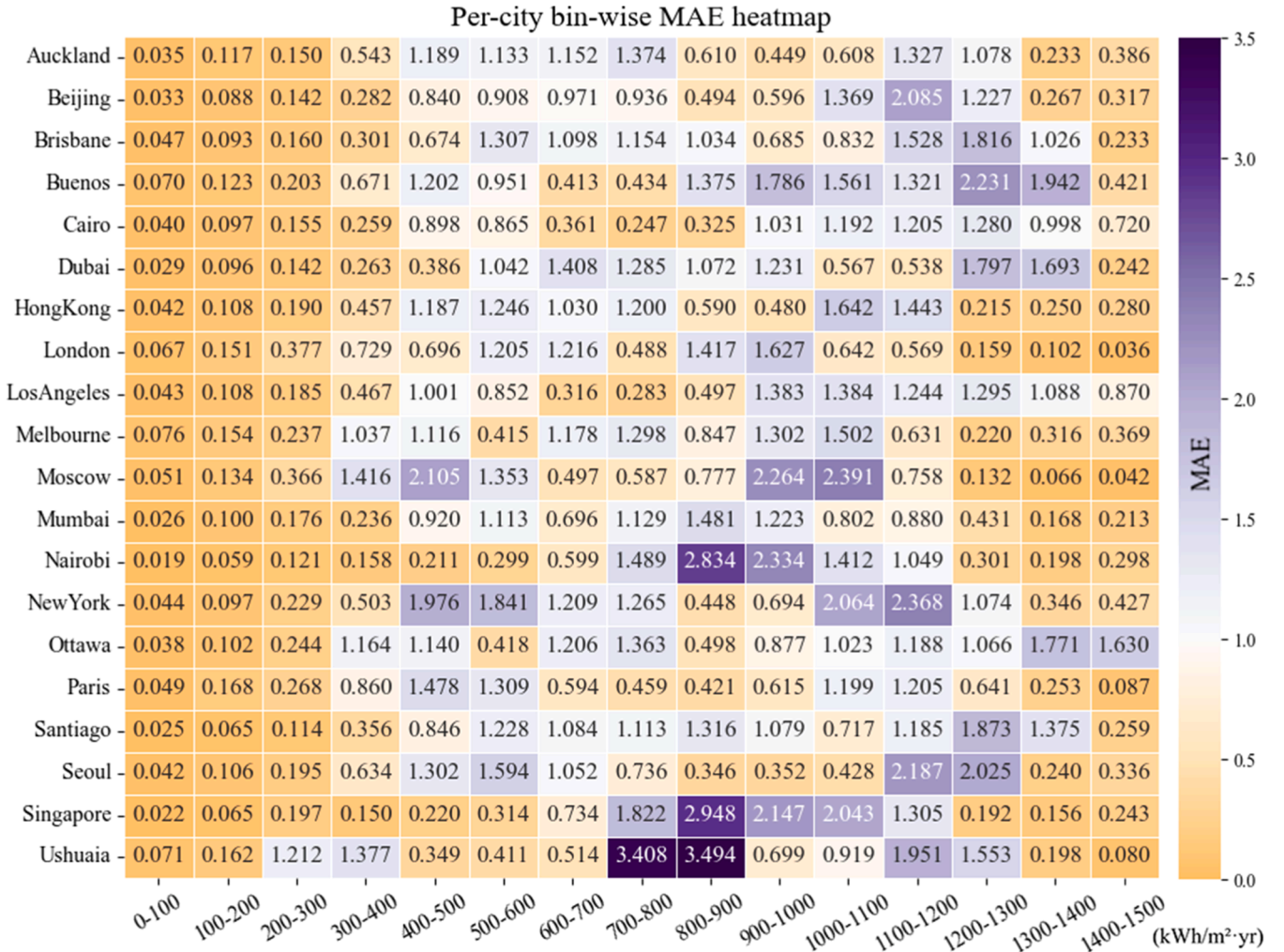
To further quantify inter-city error distributions, a bin-wise mean absolute error (MAE) was computed over annual solar radiation. As shown in Fig. 10, the model demonstrates high stability in the low-radiation range (approximately 0–700 kWh/m<sup>2</sup>·yr), while exhibiting a mild distribution bias in the medium-to-high radiation range ( $\geq 700$  kWh/m<sup>2</sup>·yr), with errors showing clear climate dependence.

A grouped analysis by absolute latitude (Table 5) indicates that the average MAE increases from low-latitude cities (0.732) to mid-latitude (0.785) and high-latitude regions (0.829). However, this trend is neither strong nor monotonic at the individual city level. Correlation analysis further reveals only a weak relationship between latitude and overall performance (Pearson  $r \approx 0.21$ ), suggesting that latitude alone is

**Table 5**

Mean absolute error (MAE) of solar radiation distribution across latitude-based city groups.

| Groups        | Num | Average MAE | Median MAE |
|---------------|-----|-------------|------------|
| low-latitude  | 4   | 0.732       | 0.725      |
| mid-latitude  | 11  | 0.785       | 0.772      |
| high-latitude | 5   | 0.829       | 0.863      |



**Fig. 10.** Heatmap of per-city bin-wise mean absolute error (MAE) between predicted and ground-truth annual solar radiation distributions, expressed in percentage points.



insufficient to explain inter-city variability. Instead, the observed differences are more plausibly attributed to the combined effects of latitude, local sky conditions, and climate-specific solar radiation characteristics.

Overall, the model is able to capture the main structure of solar radiation distribution, but its accuracy decreases under certain climatic conditions.

#### 4.6. 2AFC user study

To evaluate the visual discriminability between model predictions and physics-based simulation results, a 2AFC experiment was conducted [71]. A total of 56 participants with relevant backgrounds in architectural design or building performance analysis were recruited, including architects, engineers, researchers, and consultants. For each participant, we recorded their professional background, approximate years of experience, and commonly used modeling or simulation tools (Table S2), allowing for a more informed interpretation of perceptual performance across different expertise profiles.

In each trial, participants were presented with two pseudo-color radiation maps corresponding to the same building, viewpoint, and climate condition: one generated by Radiance simulation and the other predicted by Sketch2Radiation. The order of presentation was randomized to avoid positional bias, and participants were asked to select the image that they believed more closely resembled the simulation result, without being informed of the ground truth source. The test samples were constructed to cover a diverse set of buildings, viewpoints, and cities, and no identical sample was repeated for the same participant, ensuring independence across trials.

Experimental results show that the participants achieved an average accuracy of 56.67% (corresponding to an average error rate of 43.33%). Compared with the 50% chance level of the 2AFC task, a one-sample two-tailed *t*-test indicates that participant performance was significantly above chance,  $t(55) = 3.905$ ,  $p = 0.00026$ , with a moderate effect size (Cohen's  $d = 0.522$ ). This suggests that, while participants are able to distinguish between the two sources to some extent, the visual differences are relatively subtle, and the predicted radiation maps are broadly consistent with the simulation outputs at a perceptual level.

In addition, to further examine whether perceptual discriminability varies across professional backgrounds, we conducted an exploratory subgroup analysis by grouping participants into architects, engineers,

researchers, officer and consultants (Table S3). The results show that the average accuracy across different groups is relatively close, ranging from 53.89% to 60.00%, with an overall mean of 56.67%. Among them, the Consultant group achieves the highest average accuracy (60.00%), followed by the Architect group (58.13%). In comparison, the Researcher group shows the lowest average accuracy (53.89%), while the Engineer (55.00%) and Officer (54.67%) groups fall within a similar range. Overall, these results suggest that professional background may have a limited influence on performance in this 2AFC task, but the magnitude of the differences remains modest.

#### 4.7. Error patterns and analysis

We analyzed error patterns using representative test cases (Fig. 11). For simple massing with clear volumetric hierarchy (e.g., single-box forms or mid-rise blocks with limited setbacks), predictions closely match the reference, with only minor deviations. In contrast, larger discrepancies are consistently observed in geometrically complex high-rise towers and on curved or irregular façades, indicating a geometry-dependent error distribution.

Errors are primarily concentrated near geometric transitions where self-shading changes abruptly, such as depth discontinuities and convex-concave junctions (e.g., corners, balconies, overhangs, and skybridges). These regions exhibit high spatial gradients in both depth and visibility, resulting in sharp radiative discontinuities that are inherently difficult to approximate with smooth function mappings. Consequently, the model becomes sensitive to slight spatial misalignment and produces mild feathering along shadow boundaries, typically observed as narrow bands of over- and underestimation while preserving the overall radiation distribution.

In complex high-rise assemblies (e.g., interlocking volumes, courtyards, deep recesses, and narrow atria), occlusion relationships become highly nonlinear. Although the model reliably reconstructs the macroscopic distribution of high- and low-radiation regions, it systematically underestimates high-frequency spatial variations and attenuates sharp local extrema. This results in low-frequency bias patches and spatial smoothing effects, particularly in deeply occluded zones. Similar behavior is observed in curved or irregular geometries, where radiation varies more continuously and shading is more sensitive to viewing direction. Additionally, dense or overlapping sketch strokes may obscure fine geometric cues, introducing ambiguity in local occlusion inference.

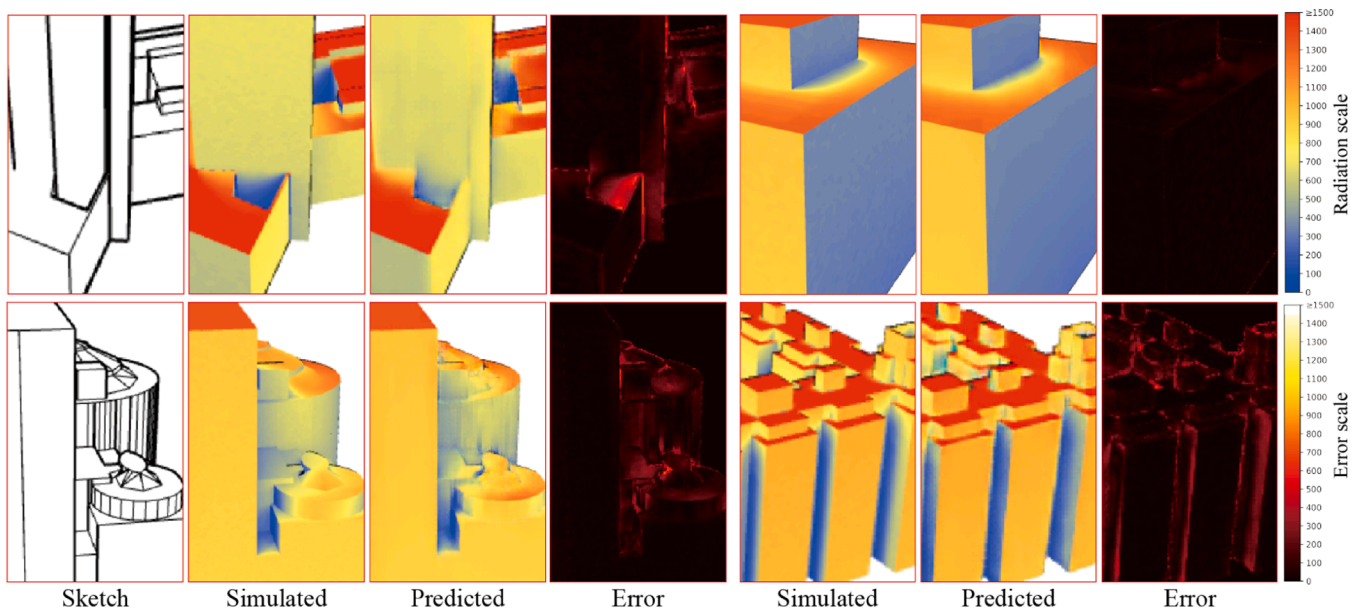


Fig. 11. Typical error patterns in simple regular massing, complex high-rise geometries and curved facades.



These effects collectively reflect the intrinsic difficulty of learning radiation fields under rapidly varying visibility conditions, where small geometric perturbations can induce large radiative changes.

From a physical perspective, sharp shadow boundaries correspond to visibility discontinuities in direct solar irradiation, where small variations in geometry or viewpoint lead to abrupt transitions in incident radiation. From a learning perspective, regression-based objectives (e.g., L1 loss) inherently favor spatial smoothness by penalizing large local deviations, introducing a bias toward low-frequency approximations and suppressing high-frequency transitions such as sharp shadow edges. This tendency is further amplified in complex geometries with dense occlusion interactions. Moreover, skip connections in U-Net architectures propagate low-level geometric features without explicitly encoding visibility or illumination constraints, which can contribute to misalignment between geometric edges and radiative discontinuities.

Several improvement directions can be considered at three levels. (i) **Architecture**: attention-modulated skip connections or boundary-aware refinement modules to improve feature selection and edge localization. (ii) **Loss design**: edge-aware, gradient-consistency, or boundary-weighted losses to enhance sensitivity to high-frequency variations. (iii) **Supervision**: auxiliary signals such as shadow masks, visibility maps, or decomposed radiation components (e.g., direct vs. diffuse) to better constrain shadow boundary learning.

Overall, Sketch2Radiation effectively captures the large-scale structure of annual façade radiation fields and reliably identifies high-PV-potential regions for concept-stage massing studies. However, it exhibits limitations in resolving sharp shadow discontinuities and fine-scale spatial variations in complex or curved geometries.

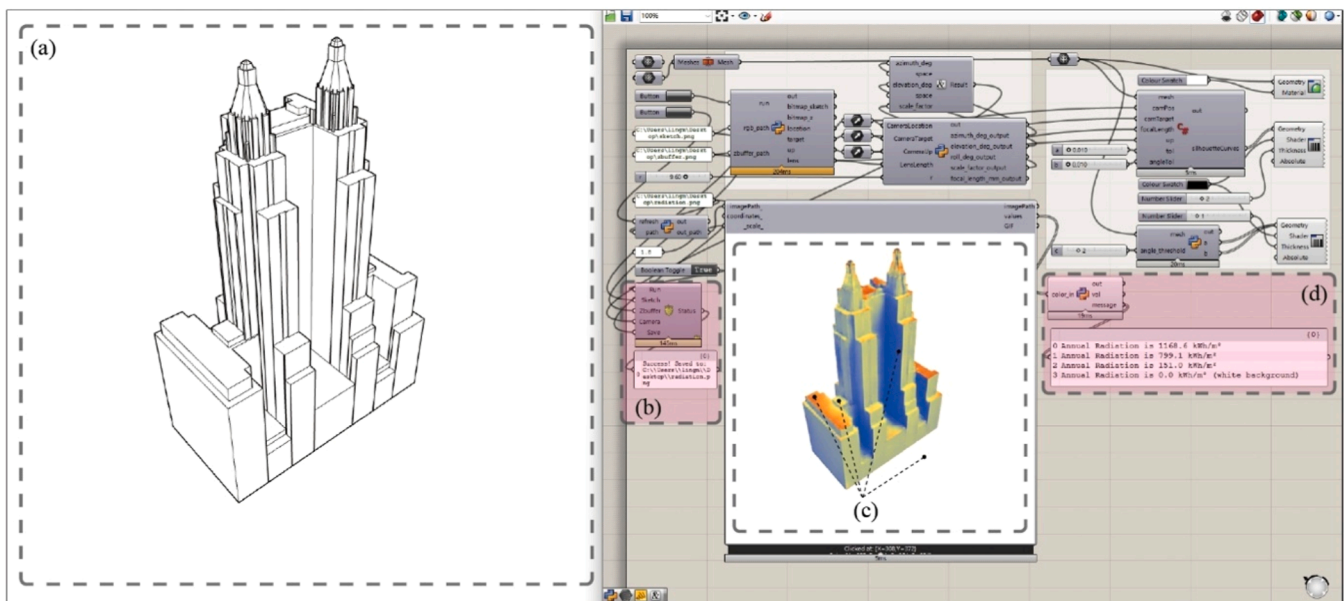
#### 4.8. Integration into parametric design workflow

To demonstrate practical viability in real design workflows, the trained Sketch2Radiation model is integrated into the Rhino and Grasshopper (Fig. 12). In this prototype, when designers manipulate parametric massing models, a Grasshopper definition automatically exports three view-aligned inputs from the active viewport: a sketch-like line drawing, a co-registered depth (Z-buffer) map, and the corresponding camera pose. These are forwarded to a GPU-backed Flask inference service, which returns the predicted radiation map for in-viewport visualization. The integration preserves designers' habitual

modeling process while delivering view-consistent solar radiation feedback directly in the working view, thereby supporting rapid iteration and side-by-side comparison of alternatives.

The current end-to-end response time is approximately 0.3 s per view, corresponding to roughly 3 FPS, which supports fast interaction and comparative exploration at the conceptual stage. Importantly, this latency is not limited by model compute. In a benchmark over 9600 test samples, pure network inference averages only  $9.1 \times 10^{-5}$  s per sample, leaving substantial headroom. Most delay arises from integration overhead, including service invocation and data transfer, file handling and disk I/O, viewport image export, preprocessing and format conversion, and repeated recomputation triggered by Grasshopper component updates during parameter edits. These observations indicate that the dominant bottleneck lies in data transfer and system-level integration overhead, rather than neural inference itself. These costs are largely implementation-dependent, indicating clear opportunities to reduce latency without changing the core model.

For engineering-grade deployment, optimization should primarily focus on shortening the data path rather than further accelerating the model itself. The main optimization pathways can be summarized as follows: (i) **Reducing data transfer overhead**. Replacing file-based handoffs with in-memory passing can eliminate disk I/O and reduce serialization overhead. (ii) **Avoiding repeated service overhead**. Maintaining a warm inference service with persistent connections avoids repeated startup and handshake costs. (iii) **Improving responsiveness under continuous edits**. Asynchronous queues and batching improve throughput and responsiveness during frequent parameter updates. (iv) **Minimizing redundant preprocessing**. Preprocessing and format conversion can be consolidated to avoid repeated operations and, where possible, moved closer to the inference backend. (v) **Shortening communication paths**. A tighter deployment form, such as a local plugin or a lightweight desktop service with direct inter-process communication to Rhino, can further reduce latency and jitter. (vi) **Avoiding unnecessary recomputation**. Caching and incremental updates can skip unnecessary exports and inference when the camera is unchanged or geometry modifications are minor, enabling smoother feedback during continuous parameter adjustment. These strategies highlight that the current latency is largely implementation-dependent and can be significantly reduced without modifying the core model architecture.



**Fig. 12.** Integration of surrogate model in Rhino via Grasshopper. (a) Designer-edited building massing in Rhino. (b) Custom Grasshopper interface for sketch and depth map generation. (c) Predicted annual solar radiation map with sample points. (d) Output radiation values at sample points.

Compared with physics-based workflows such as Ladybug, where per-view runtimes typically grow with model complexity, mesh subdivision and sensor density, Sketch2Radiation is driven by fixed-resolution image inference and is largely insensitive to meshing choices and geometric detail. This yields stable response times even for complex concept models. The modular integration architecture also facilitates future extensions to additional performance indicators, such as daylight or thermal comfort, and to optimization-driven workflows, providing a practical foundation for low-friction, view-aligned performance feedback in early-stage design.

## 5. Discussions

### 5.1. Sketch-based view-aligned solar radiation surrogate

Methodologically, Sketch2Radiation reframes the conventional workflow of explicit 3D modeling followed by BPS simulation and post-processing as a view-aligned surrogate prediction task. Rather than reconstructing full geometry, it learns a direct mapping from sketch-like design inputs to facade-scale annual radiation fields, producing outputs in the same view space where designers sketch, inspect, and iterate. This formulation complements urban-scale radiation surrogates that operate on volumetric, point-based, or graph-based scene representations. By targeting early-stage design interaction, Sketch2Radiation enables rapid, visually intuitive assessment and comparison of the facades that are visible in the active working view.

### 5.2. Intended use cases and role in early-stage design

In practice, Sketch2Radiation is intended as a concept-stage, view-aligned surrogate for designers working in early digital modeling environments such as Rhino, Revit, or SketchUp. It provides fast, view-consistent facade radiation feedback to support screening and prioritization of candidate facades, ranking of competing massing/facade variants under consistent assumptions, and comparative reasoning about how geometric edits (e.g., setbacks, overhangs, articulation) shift the spatial distribution of annual radiation during iteration. Accordingly, the framework should be interpreted as a decision-support tool for early exploration rather than a final validation engine. When a project reaches phases requiring regulatory compliance, permitting, procurement decisions, or bankable financial assessment (including long-horizon yield forecasting), project-specific context-aware and high-fidelity simulation workflows (e.g., Radiance) remain essential. Used in this way, Sketch2Radiation can shorten early evaluation cycles and help focus detailed simulation effort on a small set of shortlisted options where rigorous verification is most valuable.

### 5.3. Pathways toward freehand sketches on paper

Although the present study uses digital sketches as inputs, the formulation is intentionally open to a natural extension toward freehand drawings on paper. Our focus on a single view is not merely a simplification, but a principled choice to stay close to the first visual articulation designers produce on paper or on screen, while establishing a reusable interface and experimental baseline for inferring facade-level radiation fields from unconstrained sketches.

However, extending the current model to hand-drawn sketches on paper encounters several physical and informational obstacles. Unlike digital input, hand-drawn sketches typically do not provide explicit depth information or explicit viewpoint parameters. Therefore, occlusion relationships, surface orientation, and viewing angles become ambiguous, making it difficult to reliably infer the geometric context required for facade radiation prediction. Furthermore, hand-drawn sketches often contain strokes that do not strictly correspond to physical edges, including symbolic or stylistic elements, which can further complicate feature interpretation. These factors suggest that the

challenge lies not only in differences in visual style but also in the lack of physically relevant information.

At least two pathways appear viable. The first is a geometry-reconstruction pathway grounded in structure and information: by exploiting perspective cues such as vanishing points, the topological and geometric relations encoded in line networks (for example, intersections, junctions, and incidence constraints), along with coarse scale references and information-theoretic priors, a freehand sketch can be lifted into an approximate massing model or a depth representation, after which Sketch2Radiation can be reused as the downstream radiation surrogate. This route decouples geometric reasoning from radiation prediction, enabling modular integration with existing sketch-based modeling tools and a gradual, engineering-friendly deployment.

The second is an end-to-end, image-to-field learning pathway: with a sufficiently large dataset of freehand sketches, corresponding massings and simulated radiation maps, a model can be trained to map 2D drawings directly to facade radiation distributions, while explicitly estimating predictive uncertainty to flag risks arising from geometric ambiguity and missing visual evidence. The main obstacles lie in the current lack of large-scale paired data, the wide stylistic diversity of hand drawings, and the inherent non-uniqueness of recovering 3D form from 2D lines; nevertheless, if established, this direction would align most closely with how design is actually generated and iterated in practice.

### 5.4. Bias of the geometry source

Although the proposed dataset expands the climate coverage to 20 cities, the building geometries are primarily sourced from the Manhattan-1000 dataset, which is dominated by high-density urban buildings with relatively regular massing. While we attempted to maintain diversity and balance in the selected building samples, the current training set may still under-represent globally diverse architectural forms, especially for certain cities—such as vernacular typologies, mixed-height and mixed-roof buildings, and non-rectilinear/curvilinear envelopes. Therefore, although our experiments demonstrate robust performance across climates within the sampled geometry range, generalization to substantially different building-form distributions should be interpreted with caution in practical applications. Future work will incorporate more diverse geometry sources and adopt cross-typology evaluation protocols to systematically assess and improve transferability.

### 5.5. Limitation: context-aware vs self-shading-only

From an information-theoretic and observability standpoint, a single-view 2D-to-2D formulation carries an inherent limitation: it cannot reliably recover the full set of contextual occluders that govern urban solar access. Surrounding buildings may sit behind the target mass, outside the camera frustum, or occupy only a handful of pixels in the current view, yet they can substantially alter sky visibility and the partitioning of direct and diffuse irradiance. As a result, even with line drawings and per-view depth cues, inferring complete context shading in a stable manner is fundamentally underdetermined.

Given these physical and informational limitations, the main dataset and model in this study are constructed under a self-shading-only assumption, where only the target building's own geometry contributes to shading. In real cities, however, shading cast by neighboring buildings can be a dominant factor for facade PV potential, especially at lower floors. To make this gap explicit and measurable, we conduct additional experiments on real urban districts by comparing annual Radiance simulations under self-shading-only and fully context-aware settings, thereby characterizing the systematic bias introduced by ignoring surrounding obstructions across different density regimes and degrees of sky openness.

Specifically, we analyze annual solar radiation for 1322 real-world

buildings across diverse urban contexts, including high-density districts in New York City, downtown and public housing estates in Hong Kong, as well as lower-density suburban areas in Berlin and Montréal. For each building, we compute the SVF as hemispherical sky visibility at a point located at the building centroid at 1.2 m above ground, within a 300 m radius neighborhood. The context-aware and self-only Radiance simulations are conducted using identical geometry, the same EPW weather file, and the same Radiance parameter settings, so the resulting discrepancy isolates the impact of surrounding-building occlusion. All statistics are aggregated at the building level. Fig. 13 visualizes the relationship between SVF and the error fraction, with point color encoding building height and point size indicating footprint area. The results show that the error has an approximately linear relationship with the SVF, with linear regression performance of  $MAE = 0.053096$ ,  $MSE = 0.004892$ ,  $RMSE = 0.069946$ , and  $R^2 = 0.877163$ . Fig. 14 further reports the error distribution across different SVF ranges.

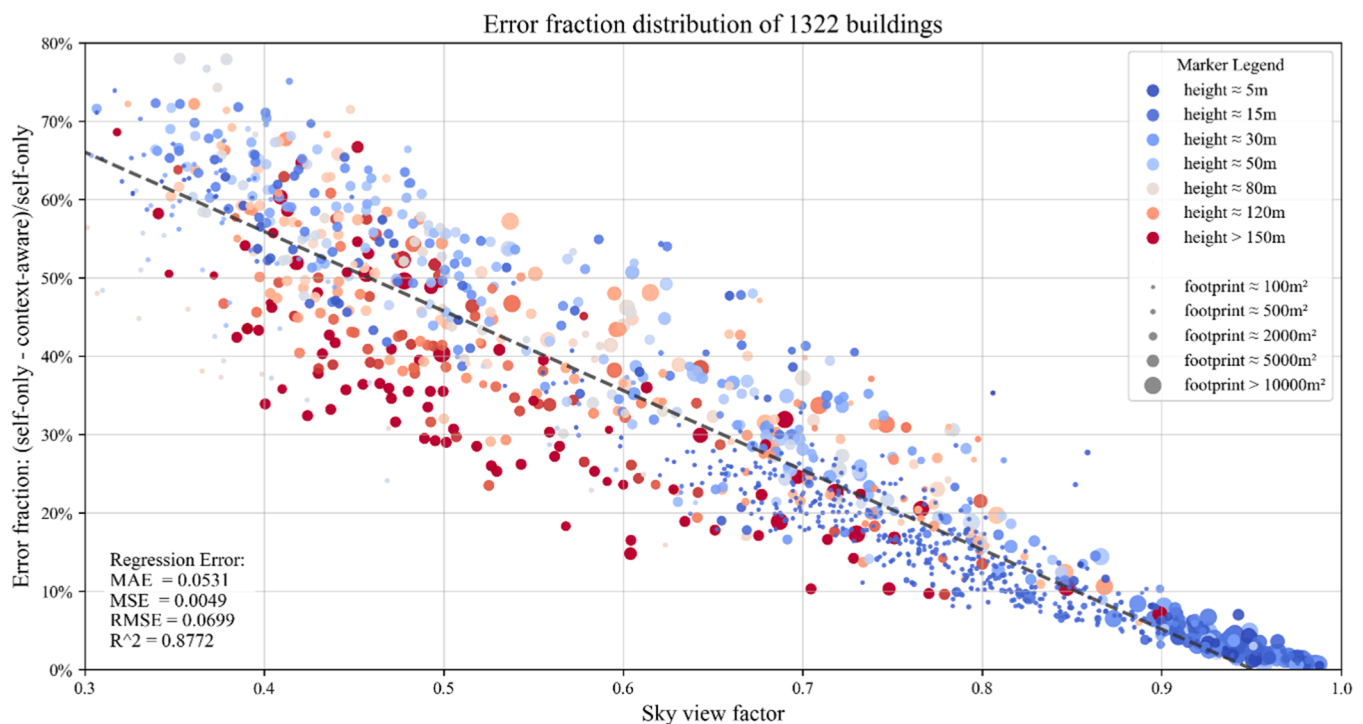
To further translate these observations into actionable guidance for design practice, SVF can be interpreted in terms of empirical confidence ranges. Based on the present dataset, when  $SVF > 0.9$ , the median error remains low ( $\approx 3.1\%$ ), and self-shading-only predictions can generally be considered reliable for rapid screening and early-stage decision-making. When SVF falls between 0.8 and 0.9, the median error increases ( $\approx 10\%$ ), but predictions remain useful for comparative evaluation and option ranking; in such cases, context-aware validation is recommended for shortlisted schemes or critical façade regions. When  $SVF \leq 0.8$ , the influence of contextual obstruction becomes more pronounced, and self-shading-only predictions should be treated as optimistic upper bounds, with high-fidelity context-aware simulation generally required before drawing performance- or investment-related conclusions. These thresholds should be interpreted as empirical guidance rather than universal criteria. While SVF provides a useful scalar indicator of contextual openness, it is not a sufficient condition: localized obstructions or nearby high-rise elements may still produce significant shading effects even under relatively high SVF conditions.

In dense urban cores, both the magnitude and variability of the bias increase substantially. In such cases, self-shading-only predictions

should be interpreted as an optimistic upper bound and used primarily for rapid screening, ranking, and comparative iteration. For practical projects embedded in existing built environments, we recommend a two-stage integration workflow: (i) apply Sketch2Radiance early to screen and prioritize façade areas and shortlist a small set of promising massing options; (ii) for shortlisted options and critical façade regions, perform context-aware high-fidelity validation (e.g., Radiance including surrounding massing) to quantify obstruction effects. Importantly, context-aware simulation validation should be conducted during the schematic development phase before drawing performance-, compliance-, or investment-related conclusions.

When context geometry is available, a straightforward enhancement is to use SVF as a correction variable, applying an empirical mapping or piecewise regression to adjust self-only predictions toward context-aware estimates at minimal additional computational cost while retaining interpretability. Looking forward, a longer-term direction is to incorporate coarse contextual descriptors directly into the data and representation, such as street-canyon morphology indicators, skyline masks, or neighborhood height statistics, so that the framework can progressively extend its reliability in high-density cities without sacrificing speed performance.

Beyond such descriptor-level extensions, these contextual signals could also be integrated more deeply into the model architecture through conditioning mechanisms, for example by injecting them at the bottleneck or intermediate feature levels using feature-wise modulation or attention-based schemes, allowing the network to adapt its internal representations to context-dependent shading patterns. A further physically grounded direction is to decompose façade radiation into a baseline self-shading component and a context-induced attenuation component, where the latter is modeled as a learned correction field conditioned on contextual priors. This formulation aligns more closely with the underlying radiative processes while preserving the efficiency and view-aligned nature of the current framework.



**Fig. 13.** Overestimation under the self-shading-only assumption compared to the context-aware case: error fraction versus sky view factor (SVF) for 1322 real-world buildings. Point color encodes building height and point size indicates footprint area; the fitted regression highlights the near-linear dependence on SVF.

Distributions of error fraction across different SVF

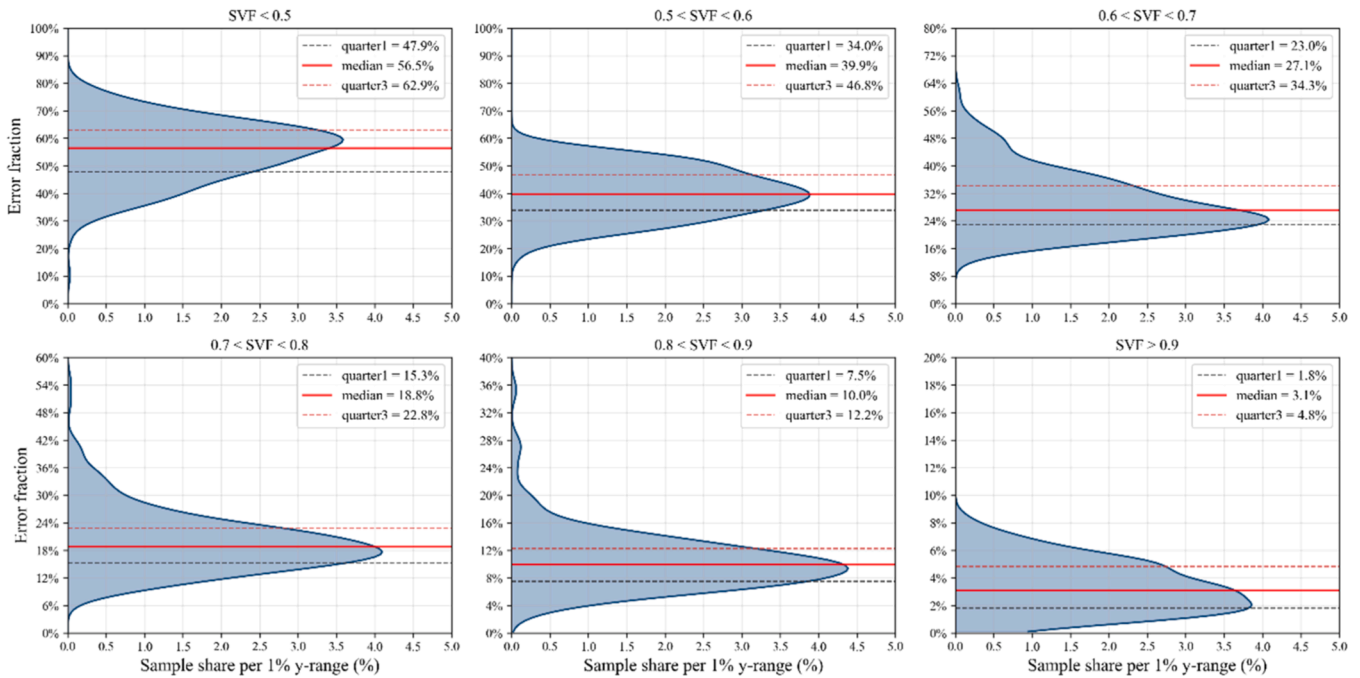


Fig. 14. Distribution of overestimation error fractions across SVF bands.

## 6. Conclusions

This paper presents Sketch2Radiation, a view-aligned surrogate framework for facade-level annual solar radiation prediction in early-stage design. Departing from conventional workflows that rely on explicit 3D reconstruction followed by building performance simulation and post-processing, Sketch2Radiation learns a direct mapping from a single design view to an annual radiation field on the building envelope. Using only a digitally generated sketch, a depth map (Z-buffer), camera pose, and a compact climate descriptor, the model produces view-consistent radiation distribution maps that can be queried at sub-second speeds, bringing performance feedback into the same visual space where designers iterate.

To support learning across both geometric diversity and climatic variability, we construct a large-scale multi-view, multi-climate dataset comprising 1000 real buildings, 24 standardized viewpoints per building, and 20 representative cities, with Radiance-based annual radiation simulations serving as supervision. Extensive evaluation demonstrates that the proposed framework achieves stable distributional fidelity at the pixel level, recovering salient spatial structures such as radiation textures, shading transitions, and the organization of high and low radiation regions. Across the full test set, Sketch2Radiation attains MSE = 0.000236, MAE = 0.002371, SSIM = 0.990219, FSIM = 0.995537 and PSNR = 39.20 dB, indicating strong agreement with high-fidelity reference simulations. Beyond pixel-wise accuracy, the model preserves design-relevant photovoltaic indicators, enabling reliable ranking and zoning of facade regions by solar potential, while a 2AFC user study further supports the perceptual plausibility and practical interpretability of the predicted radiation maps in a design context.

At the same time, this study clarifies the current scope and boundary conditions of the approach. We outline two promising directions toward freehand sketch support, namely a geometry-reconstruction route that leverages the topological and geometric cues embedded in line networks, and an end-to-end image learning route trained on large-scale triads of freehand sketches, massings, and radiation maps with explicit uncertainty estimation. We also make explicit the implications of the self-shading-only assumption adopted in the main dataset and model. A

practical next step is to introduce SVF or other lightweight context descriptors as correction variables, bridging self-only predictions toward context-aware estimates with minimal computational overhead and clear interpretability.

Overall, this study shifts solar radiation assessment from simulation-based pipelines to a view-aligned, in-workflow paradigm, enabling rapid exploration, comparison and early decision-making. Beyond providing a surrogate benchmark and a large-scale dataset, this study establishes a methodological foundation for future extensions to freehand-only sketch inputs, richer urban context modeling, and broader cross-climate generalization, advancing solar-informed architectural design from post hoc evaluation toward interactive and scalable performance guidance.

## CRedit authorship contribution statement

**Ling Mao:** Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Changying Xiang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.buildenv.2026.114725](https://doi.org/10.1016/j.buildenv.2026.114725).

## Data availability

Data will be made available upon reasonable request.



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